Paleomagnetism of the 1180 Ma Grenvillian Umfraville gabbro, Ontario

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The 1180 ± 20 Ma Umfraville gabbro is a slightly metamorphosed stock in the Grenville Province. It was sampled at 17 sites (183 specimens). After alternating field and thermal demagnetization the gabbro retains the same very stable remanent magnetization of primary origin both before and after screening. The screened AF direction from 10 sites (96 specimens) gives a pole position of 166°E, 11°S (δp = 7°, δm = 10°). This position is ≈13° southeast of that originally reported for the gabbro by Hood. This pole position either does not fit or necessitates modifications to the four hypotheses that have been invoked to explain Grenville paleomagnetic data.

Le gabbro d'Umfraville datant de 1180 ± 20 Ma est un massif légèrement métamorphisé de la province de Grenville. On l'a échantillonné à 17 localités (183 spécimens). Après désaimantation thermique et dans un champ alternatif, le gabbro retient la même aimantation rémanente très stable d'origine primaire avant et après traitement. La direction isolée par champ alternatif à 10 des sites (96 spécimens) donne un pôle dont la position est 166°E, 11°S (δp = 7°, δm = 10°). Cette position est à environ 13° au sud-est de celle qui a été rapportée par Hood pour ce gabbro. On doit considérer cette position du pôle comme aberrante ou encore on doit modifier les quatre hypothèses qu'on a invoquées pour expliquer les données paléomagnétiques du Grenville.
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Introduction

The Umfraville gabbro is located about 20 km south of Bancroft, Ontario, in the Hastings Basin of the Grenville Province. Hood (1958) selected the gabbro for one of the earliest paleomagnetic studies in Canada because it is virtually unmetamorphosed. The use of alternating field (AF) demagnetization was just being developed at that time. Hood (1958) did an AF cleaning test on one specimen from each of 12 hand samples collected from along the old Hastings Road (Fig. 1), and rejected 3 samples as magnetically unstable. Cox and Doell (1960) calculated a pole position of 168°E, 1°S using the natural remanent magnetization (NRM) directions of 58 specimens from the remaining 9 samples. This result has been often disregarded because of the lack of AF cleaning on all specimens. Nevertheless the Umfraville gabbro is one of the most likely units in the Grenville Province to have retained a primary thermoremanent magnetization (TRM), which bears directly on arguments concerning the possible translation or rotation of the Grenville orogen relative to the rest of the Canadian Precambrian Shield.

Geology

The Umfraville gabbro was first mapped by Adams and Barlow (1910) on a regional scale. Recently it has been mapped at 1 in. : ¼ mile as part of four map-areas by Hewitt and James (1956), Hewitt (1959, 1962), and Lumbers (1969).

The host rocks in the Hastings Basin for the Umfraville gabbro are strata of the Hermon and Mayo Groups (Lumbers 1967) (Fig. 1) that have been regionally metamorphosed to greenschist and amphibolite facies. The sequence grades from the basal mafic metavolcanics of the Tudor formation (lower Hermon Group), upwards through felsic metavolcanics of the Oak Lake formation (middle Hermon Group), through clastic siliceous metasediments of the Vansickle formation (upper Hermon Group), which include the Turriff metavolcanic sequence, to the uppermost carbonate metasedimentary strata of the Dungannon formation (Mayo Group), which also includes the Burnt Lake metavolcanic sequence. The basal Tudor metavolcanics were extruded about 1310 ± 15 Ma ago based on U–Pb radiometric ages for zircons from two metarhyolite specimens (Silver and Lumbers 1966).

In a major episode of intrusive activity a biotite–diorite series of plutons was emplaced some 1250 ± 25 Ma ago (Lumbers 1967, 1969). These plutons include an epizonal albite granite and syenite group associated with a culmination of regional metamorphism and deformation, and slightly younger mesozonal diorite and trondhjemite – sodic granite groups.

Following emplacement of a few minor mafic sills and dikes, a group of massive gabbro, diorite, and nepheline syenite stocks was emplaced about 1180 ± 20 Ma ago based on U–Pb dates on zircons by

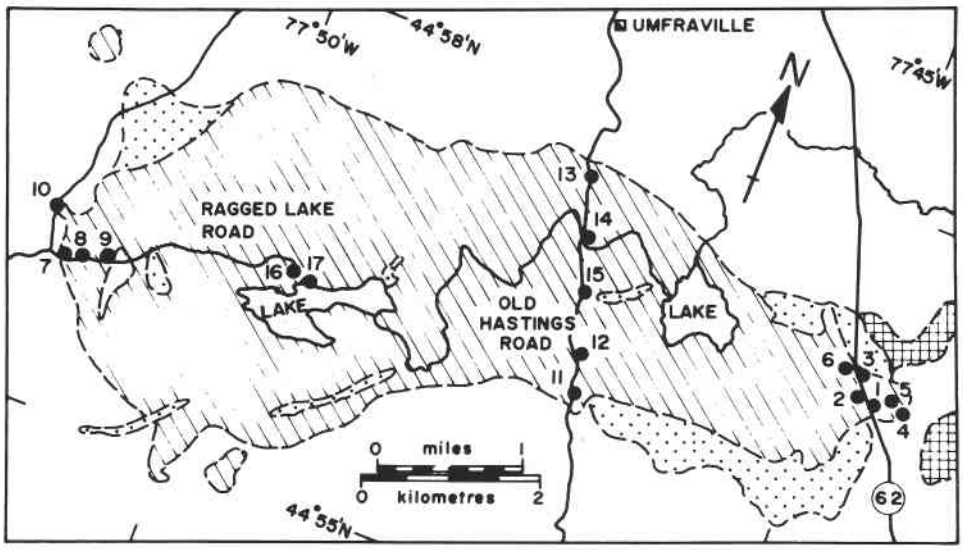


FIG. 1. Sampling sites and geology of the Umfraville gabbro. The units in order of decreasing age are: the Hermon and Mayo metavolcanics and metasediments (no pattern); the sodic intrusive trondhjemite (crossed lines); the Umfraville gabbro and related stocks (diagonal lines); and the late potassic syenite (stipple).

Silver from the syenite phase (S. B. Lumbers, personal communication, 1978). The massive Umfraville gabbro is about 10 km long by 4 km wide. It has minor zones of olivine gabbro, anorthositic gabbro, and diorite. Late phases of potassic syenite and pyroxenitic monzonite occur around its perimeter and in minor cross-cutting dikes. Titanomagnetite is the only significant magnetic mineral and it results in a 4000 nT[1] positive aeromagnetic anomaly (GSC 1959). If present, regional metamorphism of the Umfraville gabbro has not reached the greenschist facies. Other stocks in the group show local alteration of their primary pyroxene and plagioclase by autometamorphism.

Elsewhere in the Hastings Basin, minor unaltered potassic granite and syenite intrusions were emplaced between 1180 and 1050 Ma ago. These intrusions have been cut by undeformed hydrothermal quartz and calcite veins dated at 1050 ± 20 Ma by Silver and Lumbers (1966). Flat-lying Ordovician sediments unconformably overlie the Precambrian rocks in isolated outliers.

Procedure

Five cores, oriented *in situ* with a sun compass, were collected at sites 1–16, and three cores at site 17 (Fig. 1, Table 1). Two specimens were cut from each core. The NRM was measured for all 166 specimens using a Schonstedt SSM-1A spinner magnetometer and has intensities between 5 and 200 A/m. One specimen with a typical NRM direction and intensity was selected for each site. They were AF step demagnetized in 4.0 kA/m[1] increments up to 79.6 kA/m using a Schonstedt GSD-1 demagnetizer and their remanence remeasured after each step. Using the paleomagnetic stability index (PSI) method of Symons and Stupavsky (1974), an AF intensity was selected for each site in which all remaining specimens were cleaned before remeasuring their remanence. Finally, an additional typical specimen was cut for each of the 17 sites and it was thermally step demagnetized at 0, 200, 400, 450, 550, 600, 650, and 700°C in a shielded non-magnetic oven and remeasured.

Analysis

The least-squares mathematical modelling technique of Stupavsky and Symons (1976) was used to analyse data from the 17 AF step demagnetized specimens. This technique computes the direction, magnitude, and AF coercivity spectrum for each of the specimen's magnetic components. Using the 'best-fit' model when goodness of fit parameters are maximized, the most stable components have mean coercivities (H_i) from 12 to >1000 kA/m (Table 1) and a mean direction of 101.0°, 54.9° (α_{95} = 12.5°, k = 7.4) (Fisher 1953). When the five directions lying outside the cone of twice the circular standard deviation are excluded, the direction is statistically the same at 100.9°, 47.2° (α_{95} = 8.8°, k = 21.2).

Next, all the AF demagnetization data were treated in a traditional way. First, all 166 AF de-

[1]SI units: 1 γ = 1 nT, 1 Oe = 0.0796 kA/m, 1 emu/cc = 10^3 A/m, and 1 mdeg/Oe = 12.57 μdeg m A^{-1}.

TABLE 1. Remanence data by sites

Site	AF step demagnetization: PSI (μdeg m A^{-1})	$H_{\frac{1}{2}}$ (kA/m)	Cleaning field (kA/m)	Mean remanence direction: No. cores accepted/ rejected	Declination, inclination (deg)	Precision parameter k	α_{95} (deg)
1	69	>1000	19.9	3/2	106.8, 26.8	63.6	15.6*
2	286	81	39.8	3/2	88.5, 37.9	9.0	43.7
3	112	125	39.8	5/0	90.4, 50.1	35.5	13.0*
4	114	49	15.9	5/0	109.4, 23.7	485.0	3.5*
5	163	138	39.8	5/0	22.4, 43.5	8.7	27.5
6	72	115	47.8	5/0	93.8, 37.8	87.8	8.2*
7	312	20	39.8	0/5			
8	98	12	15.9	5/0	101.8, 51.5	69.6	9.2*
9	97	>1000	67.6	5/0	108.8, 45.1	309.0	4.4*
10	1610	115	19.9	4/1	29.9, 80.0	1.4	161
11	819	77	35.8	5/0	113.9, 61.9	127.0	6.8*
12	283	60	39.8	4/1	104.4, 61.2	2.0	77.1
13	274	56	19.9	4/1	107.9, 61.9	3.3	60.6
14	200	29	55.7	4/1	83.9, 47.7	14.8	24.7*
15	73	18	59.7	5/0	92.6, 69.1	3.1	52.4
16	146	30	59.7	5/0	102.1, 45.7	11.7	23.4*
17	21	35	23.9	3/0	91.0, 51.4	80.0	13.9*
Mean†				10 sites	100.4, 44.6	35.9	8.2

*Sites included in mean.
†Mean site location is 77.79°W, 44.94°N.

magnetized specimen directions were combined to give the site mean directions. These were averaged to give a unit mean direction of 95.8°, 51.2° (α_{95} = 13.1°, k = 8.4). Next these data were screened by: (a) rejecting core mean directions in which the two specimen directions diverge by more than an arbitrary 28°, (b) rejecting site mean directions represented by only one or two core means, and (c) rejecting site means with $\alpha_{95} > 25°$. Individual cores at sites 7, 10, 31, and 14 gave antiparallel directions and these were reversed in the calculation process. The 10 accepted site mean directions give a unit mean direction for the Umfraville gabbro of 100.4°, 44.6° (α_{95} = 8.2°, k = 35.9) (Fig. 2).

Nearly all of the 17 thermal step demagnetization curves are 'knee shaped' (Fig. 3). The specimens from sites 1–9 and 15–17 retain their stable direction to above 550°C at which temperature they still possess a median value of 69% of their NRM intensity. The specimens from sites 11 and 14 gave stable directions to above 500°C, site 10 to above 400°C, and sites 12 and 13 proved unstable. The specimens from sites 7, 10, and 13 gave antiparallel directions. At 650°C the directions are scattered thoroughly with specimens from sites 1, 3–9, 11, 12, 16, and 17 possessing a median value of <1% of their NRM intensity. Site 2 retains its stable direction and 9% of its NRM intensity at 650°C and site 13 moves to a stable antiparallel direction with 5% of its NRM intensity at 600 and 650°C. Hematite must carry a significant stable remanence component in these

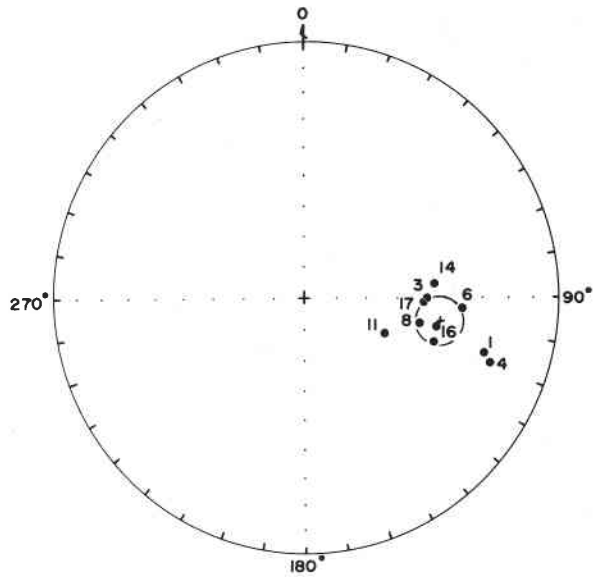


FIG. 2. Site mean remanence directions after AF demagnetization plotted on an equal-area stereonet. All directions are positive or downwardly directed. The mean direction is shown by a cross surrounded by its cone of 95% confidence.

two specimens. Sites 14 and 15 are unstable above 600°C because of changes in their ferromagnetic minerals. The thermal mean direction of 85.2°, 47.1° (α_{95} = 12.2°, k = 10.1) was computed from the specimen directions for sites 10 at 400°C, 11 and 14 at 500°C, 1–9 and 15–17 at 550°C, and 13 at 650°C with antiparallel directions reversed. When the

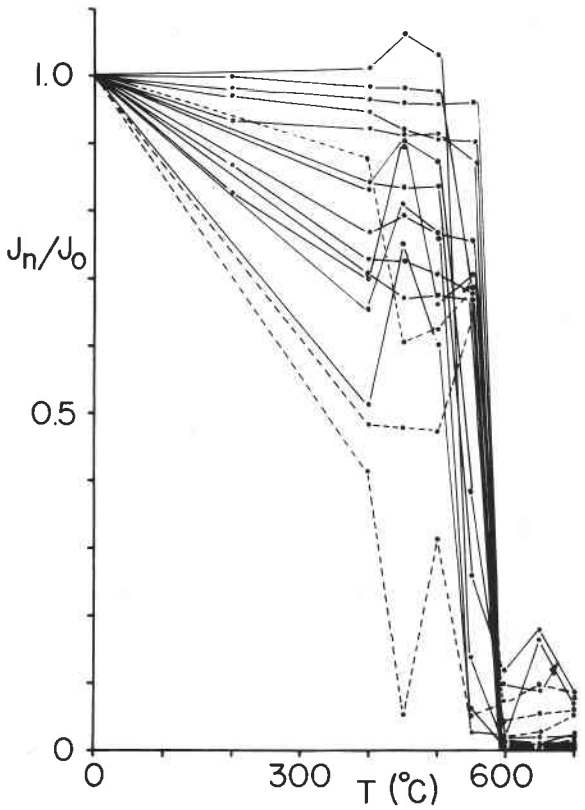


FIG. 3. Thermal demagnetization curves showing the ratio of the measured to original remanence intensity (J_n/J_o) against the cleaning temperature T in degrees Celsius. Excluding unstable site 13, all remaining 16 site specimens are shown. Sites undergoing a remanence reversal are shown by a dashed line.

three directions lying outside two times the circular standard deviation are excluded, the direction is 93.3°, 44.1° ($\alpha_{95} = 9.2°$, $k = 21.1$).

The important consequence of these various forms of analysis is that they all isolate the same stable remanence direction (Fig. 4). The remanence resides dominantly in relatively pure magnetite with Curie temperatures above 550°C with minor components in hematite with Néel temperatures above 650°C. Following Pullaiah *et al.* (1975), this stable remanence could only be formed as a primary TRM or in regionally metamorphosed rocks of upper greenschist or higher grade. Because the Umfraville gabbro has not been metamorphosed or altered, it is most unlikely to have a post-emplacement chemical remanent or metamorphic thermal remanent magnetization. Hence, its remanence is thought to be a primary TRM. Further, the gabbro has a narrow well-defined contact metamorphic aureole indicating emplacement into cool metamorphosed host rocks

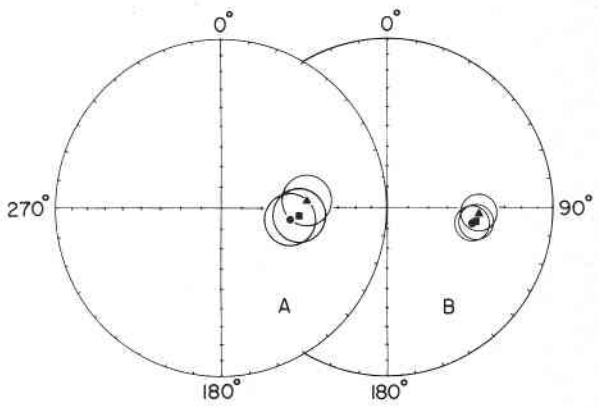


FIG. 4. Unit mean directions for the Umfraville gabbro before (A) and after (B) screening based on AF step demagnetization component analysis (circle), on the AF demagnetization of all specimens (square), and, on the thermal step demagnetization high temperature directions (triangle) with their radii of 95% confidence.

and, therefore, a prolonged cooling history of tens of millions of years is unlikely.

Discussion

The Umfraville gabbro pole of 166°E, 11°S (δp = 7°, δm = 10°) is calculated from the screened AF demagnetization direction. This pole is ≈13° southeast of the original pole (Hood 1958; Cox and Doell 1960), and the difference likely reflects the larger data base and the removal of some Recent viscous remanence from this pole. The Boulter, Thanet, and Tudor metagabbros outcrop 25 km northeast, 10 km south, and 25 km south of the Umfraville gabbro, respectively, and were emplaced at the same time. Their poles of 157°W, 42°N, 157°E, 28°S (Hood 1958; Cox and Doell 1960), and 137°E, 43°N (Palmer and Carmichael 1973), respectively, diverge markedly from the Umfraville pole. The most plausible explanation for the divergence is that the metagabbros have a secondary magnetization acquired during metamorphism. Palmer and Carmichael (1973) describe the alteration in the Tudor metagabbro. Also the Boulter and Thanet poles are based on only eight and six hand samples, respectively.

Several apparent polar wander (APW) paths and their requisite tectonic models have been proposed to explain the relationship between the Grenville Province poles and the poles from the remaining Precambrian Shield (Laurentia) of North America. This discussion is limited to 'fitting' a primary 1180 ± 20 Ma Umfraville pole into these models.

The first model assumes that the Grenville has been solidly fixed to Laurentia to form one plate, and that the APW path contains several loops or

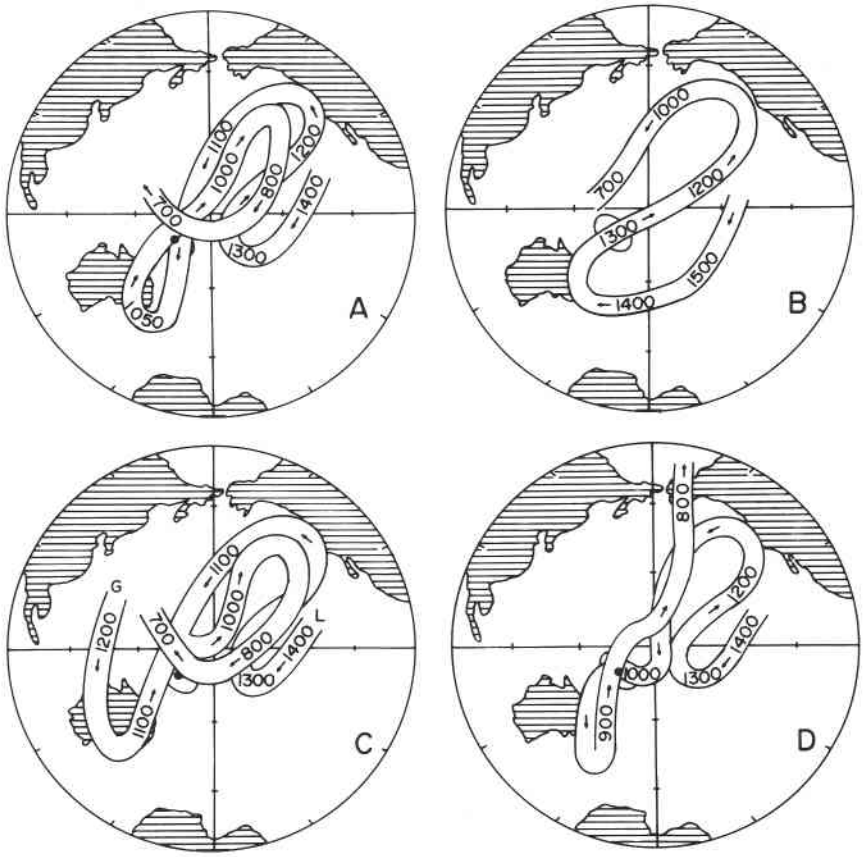


FIG. 5. Apparent polar wander curves for North America with ages in Ma based on: (a) model one assuming rapid polar wander and taken from Irving *et al.* (1974*b*); (b) model two assuming rotation of the Grenville and taken from Fahrig *et al.* (1974); (c) model three assuming the Grenville (G) and Laurentia (L) plates moved independently and taken from Irving *et al.* (1974*b*); and (d) model four assuming metamorphic thermochron poles and taken from McWilliams and Dunlop (1978).

tracks in the 900–1300 Ma interval (Irving *et al.* 1974*b*). This model requires APW to have been several times more rapid than known Phanerozoic rates. To fit this model (Fig. 5A) either the Umfraville gabbro should have an age of ≈1080 or ≈1020 Ma, or a significant extra loop is required in the APW path.

The second model assumes that the Grenville has been attached to Laurentia and rotated clockwise ≈30 ± 5° around a vertical axis with respect to Laurentia (Roy and Fahrig 1973; Fahrig *et al.* 1974; Fahrig and Jones 1976). To fit the Fahrig *et al.* (1974) model (Fig. 5B) the Umfraville gabbro should have an age of ≈1300 Ma. Alternatively, the Umfraville gabbro would have had to have been rotated ≈30° west-northwest around a horizontal north-northeast axis to bring its pole into agreement with their 1180 Ma APW path position. Contact attitudes, primary foliations where present in the massive Umfraville gabbro, and distribution of phases all suggest that the pluton was emplaced as a vertical diapir and has not been rotated.

The third model assumes rapid, large scale (>100°) translation of the Grenville plate with respect to the Laurentia plate (Irving *et al.* 1972; Palmer and Carmichael 1973; Irving *et al.* 1974*b*; Irving and Lapointe 1975). Irving *et al.* (1974*b*) use Hood's (1958) Umfraville pole with an assigned age of ≈1020 Ma to make it fit this model (Fig. 5C), but curiously they do not use his Boulter or Thanet poles. A plausible alternative would be to loop their Grenville Track back so that the 1020 and 1180 Ma APW path points are co-incident. If true, then the Umfraville gabbro was emplaced when the Grenville and Laurentia were separating and ≈70° apart. This model then requires a rather rapid 7 cm year^{-1} closure between the plates from ≈1150 to 1050 Ma. It also requires the defining of a plate suture within the Grenville near the Grenville Front for which there is little evidence as yet (McWilliams and Dunlop 1978).

The fourth model assumes that the Grenville and Laurentia formed one plate with the Grenville having reset magnetizations. In this thermochron

model, the Grenville poles were set during post-metamorphic cooling between 1050 and 800 Ma about co-incident with closure of the K–Ar radioactive decay systems (Stewart and Irving 1974; McWilliams and Dunlop 1975, 1978; Buchan and Dunlop 1976). Palmer *et al.* (1977) have shown that dikes crossing the Grenville Front into the Grenville have reset magnetizations. To fit the McWilliams and Dunlop (1978) model (Fig. 5D), the Umfraville gabbro would have had to acquire its magnetization either ≈1000 Ma ago at the start of cooling or ≈875 Ma ago near the end of cooling. They consider that the Seal–Croteau, Michael, and Mealy Mountain plutons from the Grenville Front in Labrador are unmetamorphosed, retain a primary magnetization of ≈1400 Ma, and hence exclude these poles from their analysis. These poles (Fahrig and Jones 1976) are ≈20° north of the Umfraville pole. Possibly the thermochron analysis is correct and the primary Umfraville pole should be similarly excluded.

The establishment of the Umfraville gabbro pole as a primary 1180 ± 20 Ma pole does not fit any of the four models particularly well. It does raise the probability that similar magnetite poles such as the Morin Complex (M-2) pole (Irving *et al.* 1974*a*) and the Whitestone anorthosite WY pole (Ueno *et al.* 1974) are primary and 'see through' the metamorphism whereas the corresponding hematite poles in these units are secondary chemical remanences produced during metamorphism. If true then Track 3 (Irving and Lapointe 1975), i.e. the 1300–1100 Ma loop in Fig. 5C, could swing further south and west to incorporate these primary ≈1200 Ma Grenville magnetite poles, move through the Logan Loop, continue south through Australia to incorporate secondary 1000–800 Ma Grenville magnetite and hematite poles, before returning northward along the Hadrynian track of Morris and Roy (1977).

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